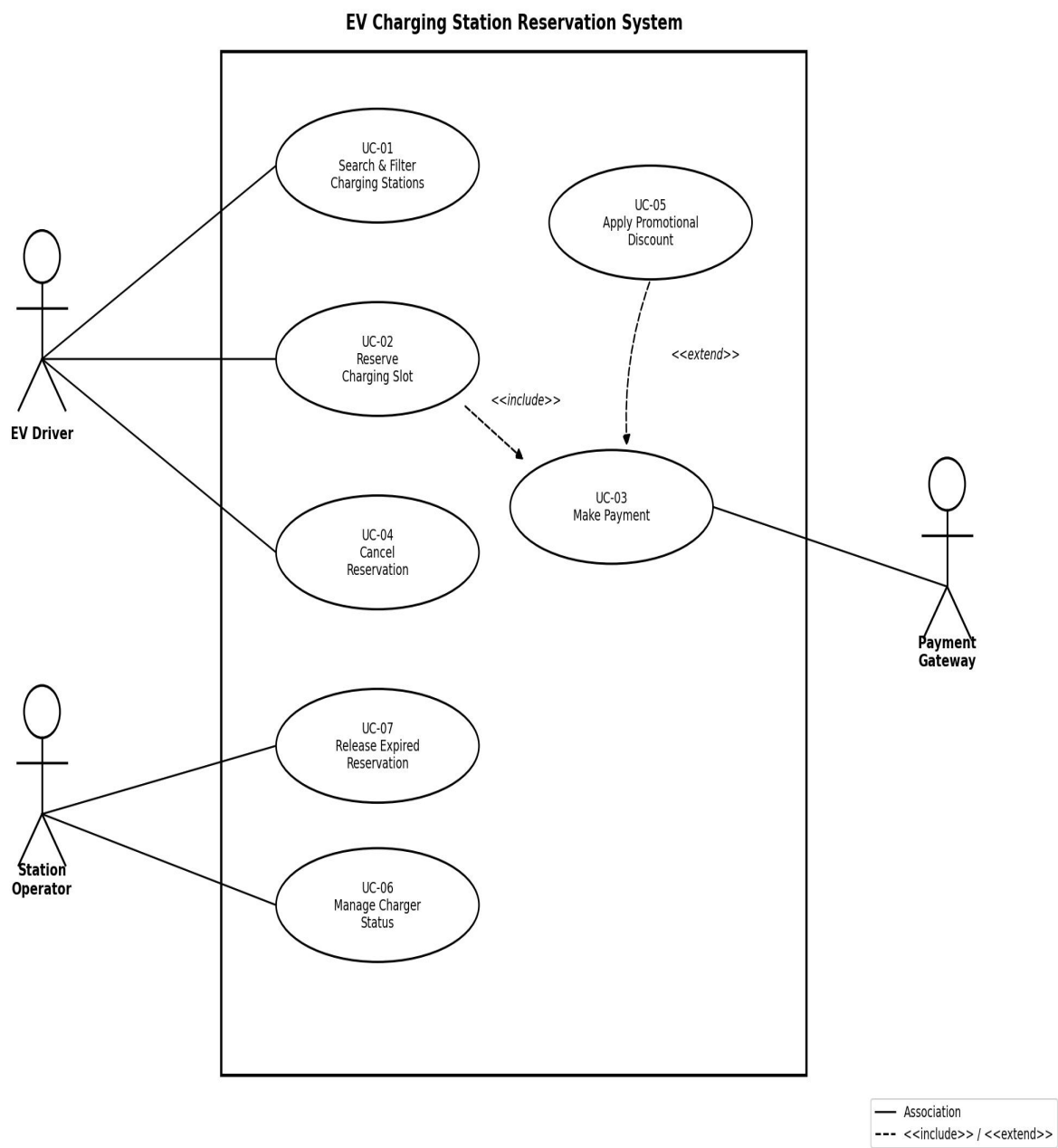


Lab 1: Requirements Engineering & UML Use-Case Modelling

1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall match the driver's vehicle connector type (CCS, CHAdeMO, or Type 2) with available station chargers and reserve a 30-minute charging window.	High	Pass: Incompatible connector plugs are hidden from the driver, and a reserved 30-minute window is created. Fail: Two vehicles are booked for the same socket concurrently, or an incompatible connector is offered.	Prevents wasted trips to incompatible chargers and guarantees the driver a usable slot on arrival.
FR-002	Functional	The system shall allow a driver to search and filter nearby charging stations by connector type, availability, and maximum distance from current location.	High	Pass: Search results returned within 3 seconds, showing only stations with at least one matching, available socket within the chosen radius. Fail: Results include stations with no matching connector or that are fully booked.	Drivers need a fast, relevant list of usable stations rather than manually checking every station.
FR-003	Functional	The system shall allow a driver to cancel a confirmed reservation up to 5 minutes before the scheduled slot start time, releasing the socket for other drivers.	Medium	Pass: Cancelling more than 5 minutes before slot start immediately frees the socket in the availability list. Fail: A cancelled slot remains shown as booked, or cancellation is accepted within the final 5 minutes without a warning.	Reduces idle 'phantom bookings' and improves socket utilisation for other drivers.
FR-004	Functional	The system shall calculate a dynamic charging rate based on time-of-day demand and station location, and display the estimated total cost before the driver confirms the reservation.	High	Pass: Estimated cost shown matches the rate table for that station/time-slot combination and is displayed before the 'Confirm' button is enabled. Fail: Reservation can be confirmed without a cost estimate being shown, or the charged amount differs from the quoted estimate.	Transparent pricing is required for driver trust and to support demand-based load balancing across stations.
FR-005	Functional	The system shall allow a Station Operator to update the real-time status of a charger (Available, Occupied, Under Maintenance) and immediately remove chargers marked 'Under Maintenance' from driver search results.	Medium	Pass: A charger marked 'Under Maintenance' disappears from driver search results within 30 seconds of the operator's update. Fail: A charger under maintenance is still offered to a driver for reservation.	Keeps station operators in control of physical hardware issues and prevents drivers from being routed to broken chargers.
NFR-001	Nonfunctional	The system shall automatically release a locked socket and apply a no-show penalty to the driver's account if they fail to plug in within 10 minutes of the reserved slot start time.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load, and sockets are auto-released and penalties logged exactly at the 10-minute mark. Fail: A socket remains locked beyond 10 minutes with no plug-in detected, or no penalty is recorded.	Protects charger availability for other drivers and discourages speculative, unused reservations.
NFR-002	Nonfunctional	The system shall support at least 500 concurrent reservation requests across all stations with a 95th-percentile response time under 2 seconds, and shall never assign the same socket to two overlapping reservations.	High	Pass: Load test with 500 concurrent users shows 95% of reservation requests completing in under 2 seconds with zero double-bookings. Fail: Response times exceed 2 seconds for more than 5% of requests, or any double-booking occurs.	The reservation step is the system's core transaction; it must stay fast and consistent even during peak commute-hour demand.

2. UML Use-Case Diagram



3. Use-Case Flow Specification

Use Case:

UC-02 — Reserve Charging Slot

Actors:

EV Driver (primary); Payment Gateway (secondary, via <<include>> Make Payment).

Preconditions:

1. The driver is logged into the app with a valid account and a saved vehicle profile that specifies the vehicle's connector type (CCS, CHAdeMO, or Type 2).
2. The driver has at least one valid payment method registered.
3. At least one station within the driver's search radius has a socket matching the vehicle's connector type.

Postconditions (Success):

1. A 30-minute charging slot is reserved and the corresponding socket is locked so no other driver can book it.
2. The driver's payment method has an authorization hold placed via UC-03 Make Payment.
3. The driver receives a confirmation notification containing the reservation code, station location, and slot start time.

Main Success Scenario:

1. Driver opens the app and selects “Find a Charging Station”.
2. System (UC-01) displays nearby stations with sockets matching the driver's connector type, along with availability and estimated cost.
3. Driver selects a station and a desired slot start time.
4. System checks real-time socket availability for that station and time.
5. System calculates the dynamic rate for the slot and displays the estimated total cost (FR-004).
6. Driver confirms the reservation request.
7. System executes UC-03 Make Payment (<<include>>) to place an authorization hold on the driver's registered payment method.
8. System locks the selected socket exclusively for the driver for a 30-minute window starting at the slot time (FR-001).
9. System generates a reservation code and sends a confirmation notification to the driver.
10. Driver arrives at the station and plugs in within 10 minutes of the slot start time.
11. Use case ends successfully.

Alternate Flow A — Socket No Longer Available (branches at step 4):

- a. System detects that the selected socket was booked by another driver a moment earlier.
- b. System displays a “Slot no longer available” message and immediately re-runs the search.
- c. Driver selects a different available station or time slot and resumes at step 3 of the main flow.

Alternate Flow B — Driver Fails to Plug In in Time (branches after step 10, ties to NFR-001):

- a. Driver does not plug in within 10 minutes of the reserved slot start time.
- b. System automatically releases the locked socket, making it available to other drivers.
- c. System applies a no-show penalty to the driver's account and sends a notification explaining the penalty.
- d. Use case ends without a completed charging session.